Applied Economics

Regression Discontinuity Design

Pello Aspuru

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Introduction

For any generic multiple linear regression model, always assume endogeneity:

$$Y_i = \beta_0 + \beta_1 X_{1,i} + u_i, \qquad i = 1, ..., n \qquad E[X_{1,i} u_i] \neq 0$$

Four **quasi-experimental methods** to *mimic random treatment assignment,* overcome endogeneity, and claim causality:

1. instrumental variables (IV) $\rightarrow$ Requires instrument Z_i
2. Panel Data Methods $\rightarrow$ Requires Panel Data
3. Diff-in-diff (DiD) $\rightarrow$ Requires Panel Data + Treatment & Control Groups
4. This session: Regression discontinuity design (RDD)

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RDD - Basic idea

When continuous variable a_i passes cutoff c, probability of treatment D_i jumps to 1.

- If observation i is above thresold $a_i > c \quad \rightarrow \quad D_i = 1$ gets treated.
- If observation i is below thresold $a_i \leq c \quad \rightarrow \quad D_i = 0$ does not get treated.

Key Assumption

Observations *around* cutoff c are **similar**, but for treatment status.

$$E[u_i D_i] = 0$$

$\rightarrow$ *As if* the treatment was randomly assigned around the cutoff!

Examples of Arbitrary Cutoffs

- Treatments for "low birth weight" newborns (>**1,500 grams**) (Almond et al., 2010)
- First-past-the-post elections (Bazzi et. al., 2025).

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RDD - Pros & Cons

Gap in outcome at the threshold $= \delta =$ *local* average treatment effect (LATE)

- RDD only tells you the impact of treatment on outcome **around the cutoff value** of the running variable. (e.g. effect of getting medical treatment at 1,500g)
- Limited **external validity**: you cannot extrapolate to the entire population. (e.g. effect of getting medical treatment at 3,000g birth weight)
- RDD is a very **data-hungry method**: only works with sufficiently large samples, because we essentially throw away observations that are not close to the cutoff.

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Example: Mortality Effect of Alcohol Consumption

RDD Framework - Example

Does alcohol consumption cause higher mortality among young adults?

Examples of endogeneity? We need a random allocation of the alcohol consumption treatment $\rightarrow$ We need quasi-random variation in alcohol consumption

In the US, alcohol consumption is prohibited before *age* reaches the cutoff $c = 21$.

Treatment status D_a is a deterministic and discontinuous function of *age*:

$$D_a = \begin{cases} 1 & \text{if } age \geq 21 \\ 0 & \text{if } age < 21 \end{cases}$$

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df <- read.csv("./AE_week11_RDD/dataRDD.csv")
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$\rightarrow$ What is the dataset's unit of observation?

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First Look at the Data

Scatterplot showcasing the three variables of interest:

- Running covariate: *age* in years $\rightarrow$ Horizontal Axis
- Treatment indicator: $D_a \rightarrow$ Color code
- Outcome variable: $Deathrate_a \rightarrow$ Vertical Axis

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rdd_plot<-
ggplot(df, aes(x = age_cell, y = death_rate_per_100T,
                        color = as.factor(treat))) +
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geom_vline(xintercept = 21, linetype = "dashed") +
theme_minimal()+theme(legend.position = "none")
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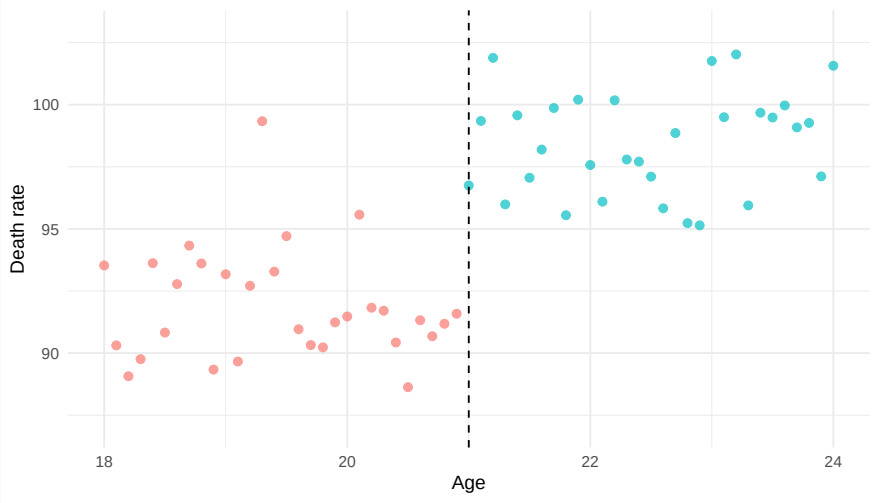
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RDD - Graphs: Running Variable and Cutoff



Estimation of the size of the gap

We can write the following simple regression model:

$$Deathrate_a = \alpha + \beta(a - cutoff) + \delta D_a + u_a$$

```
df$age_centered <- df$age_cell - 21 # New variable centered at cutoff.
rdd_lm <- lm(death_rate_per_100T ~ age_centered + treat, data=df)
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- $Deathrate_a$ death rate at age a, per 100,000 people.
- $(a - cutoff)$ age in years, centered at the cutoff.
- D_a treatment indicator.

$\rightarrow$ δ captures the "jump" in death rates above and below the 21 year old threshold.

RDD exploits a discontinuity at $a = 21$ in the conditional expectation function:

$$\delta = \underbrace{\lim_{a \to 21^+} \mathbb{E}[deathrate_a | a = c]}_{\alpha + \delta} - \underbrace{\lim_{a \to 21^-} \mathbb{E}[deathrate_a | a = c]}_{\alpha}$$

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Plotting the Regression

Generate the fitted values from the regression:

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df$fit_common_slope <- predict(rdd_lm)
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Add the fit line to the previous plot:

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Plotting the Regression

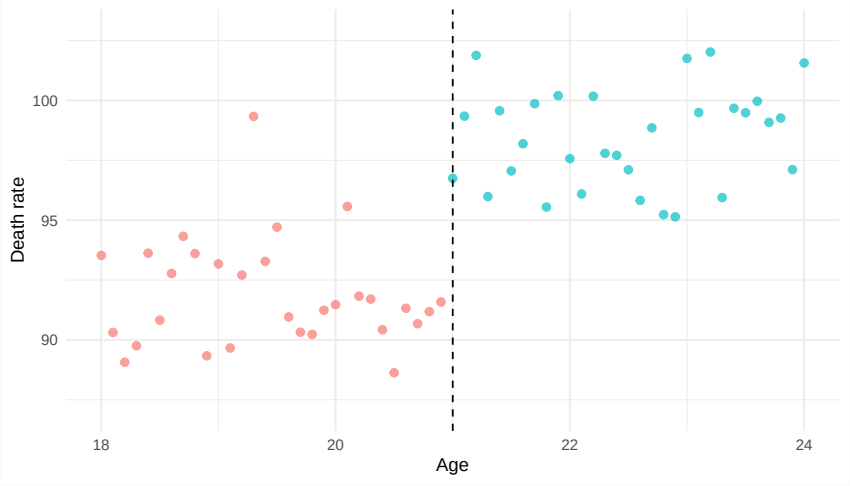
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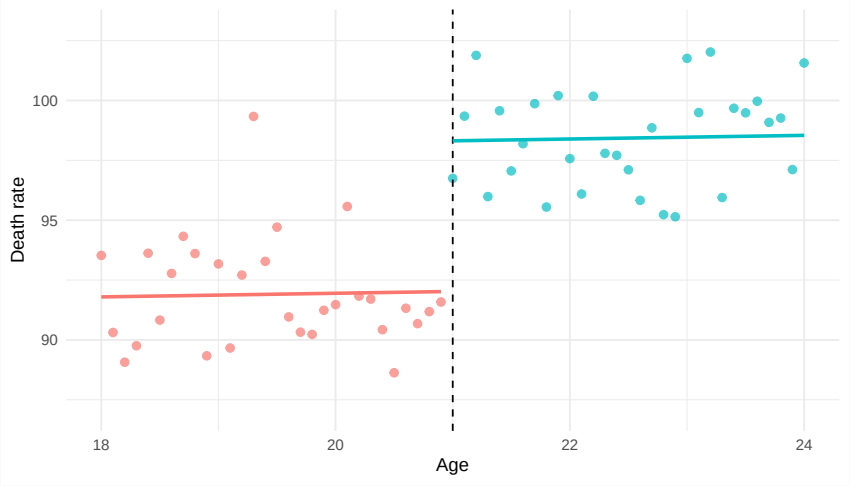
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Visualizing the Regression Discontinuity



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Identification and Sources of Bias

Identifying Assumption

Key assumption: Potential outcomes are smooth at the threshold. In other words, in the absence of treatment, there would be no discontinuity.

- Units around the cutoff are equivalent but for treatment assignment.
- The only thing that changes at the cutoff must be the treatment status

$$E[u_i|D_i] = 0$$

$\rightarrow$ Is this assumption testable?

Placebo Test

- Check if other control covariates X_a vary at the cutoff. Which
- If covariates are different above/beyond discontinuity... is it good or bad?

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Running Covariate Manipulation

Running Covariate (age_i) + Threshold ($age_i = 21$) $\rightarrow$ Treatment allocation (D_i)

- Sometimes, units can manipulate their running covariate, decide their treatment.

$\rightarrow$ Treatment no longer randomly assigned around the cutoff!

Example: food labelling policies. Products with certain calories must display a warning.

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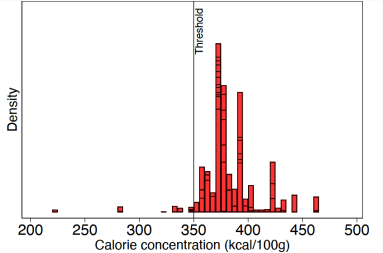
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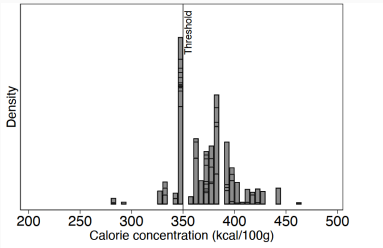
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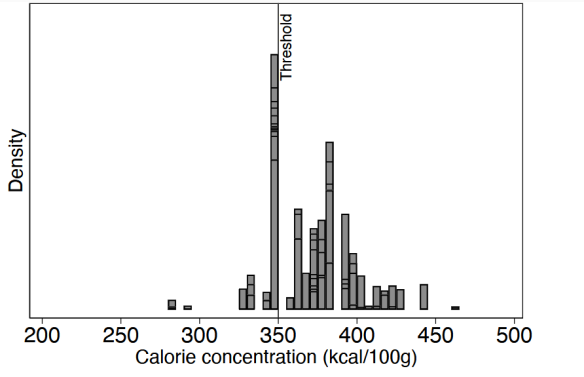
(a) Before Policy



(b) After Policy

Checking for Manipulation: Bunching

Manipulation of the running variable around the cutoff often generates a big cluster of observations to one side of the cutoff (bunching) and a gap on the other side



Another Issue: Noncompliance

Noncompliance

The running variable does not fully determine assignment to treatment. At the cutoff, your chance of treatment *increases* (jumps upward), but not from 0 to 1.

$\rightarrow$ **Fuzzy RDD** (instead of sharp RDD)

We will not cover the fuzzy RDD in detail, just know that the problem of imperfect determination of allocation to treatment can still be solved.

Coming Up

So far, we have discussed the basics of RDD. We will now focus on three practical issues when implementing RDD:

1. Functional form of the running covariate.
2. Bandwidth choice.
3. Kernel choice.

Functional Form

Fitting the Running Covariate

We assumed a linear relationship between the running covariate and outcome:

$$Deathrate_a = \alpha + \beta a + \delta D_a + u_a$$

What about trying more flexible specifications? Tradeoff:

- insufficiently flexible: risk of mistaking nonlinearity for treatment effect δ.
- overly flexible: reduces precision and runs the risk of overfitting.
- essential to visualise the data to choose an appropriate functional form!

The solution: Try different specifications!

1. linear specification with different slopes above and below the cutoff
2. Quadratic Polynomial specification

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Varying Slopes

Linear functional form, with different slopes β, γ on either side of the cutoff.

$$Y = \alpha + \beta(a - cutoff) + \gamma(a - cutoff) \cdot D_a + \delta D_a + u_a$$

```
rdd_2slopes <- lm(death_rate_per_100T ~ age_centered*treat, data=df)
coeftest(rdd_2slopes, vcov. = vcovHC(rdd_2slopes, "HC0"))
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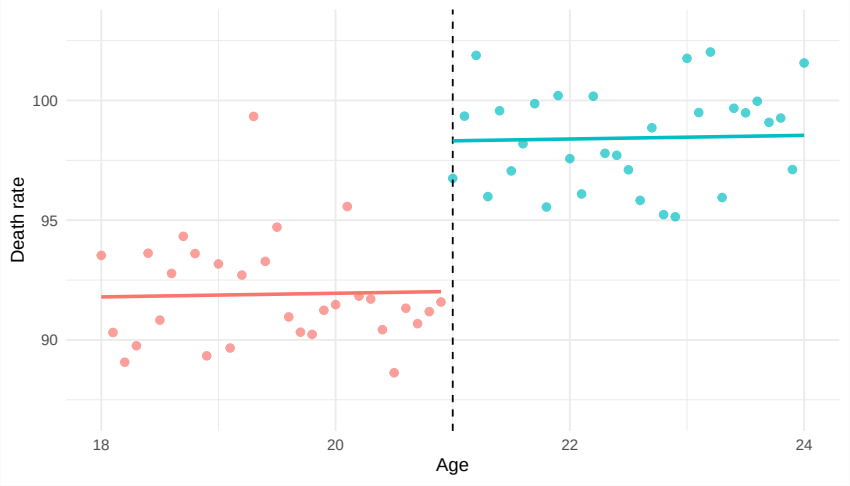
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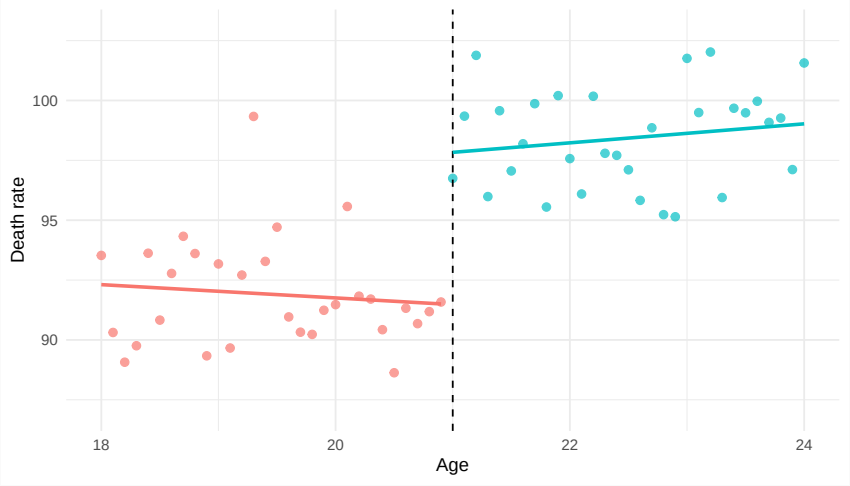
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Varying Slopes - Graphical Results



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Quadratic Polynomials

Further improve fit by adding quadratic terms β_2, γ_2 below and above cutoff:

$$\begin{aligned} Y =& \alpha + \beta_1(a - cutoff) + \beta_2(a - cutoff)^2 + \\ & \gamma_1(a - cutoff) \cdot D_a + \gamma_2(a - cutoff)^2 \cdot D_a + \delta D_a + u_a \end{aligned}$$

```
df$age_centered_sq <- (df$age_centered)^2 # new variable

rdd_quadratic <- lm(death_rate_per_100T ~ age_centered*treat +
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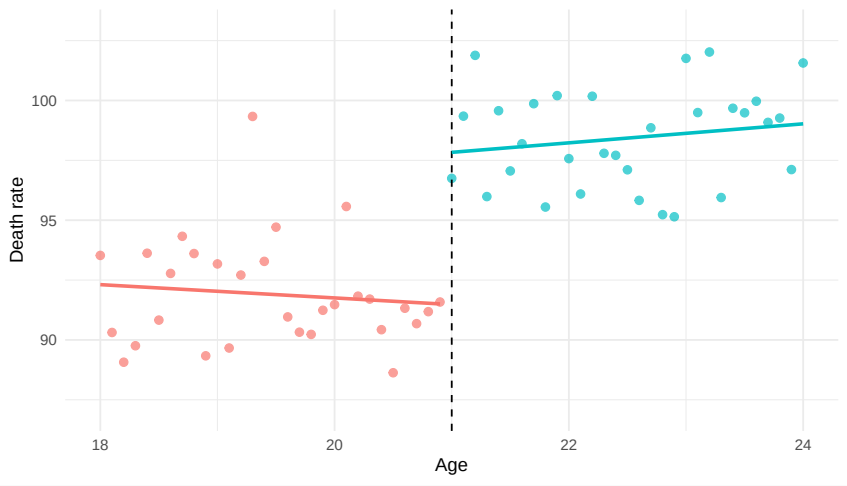
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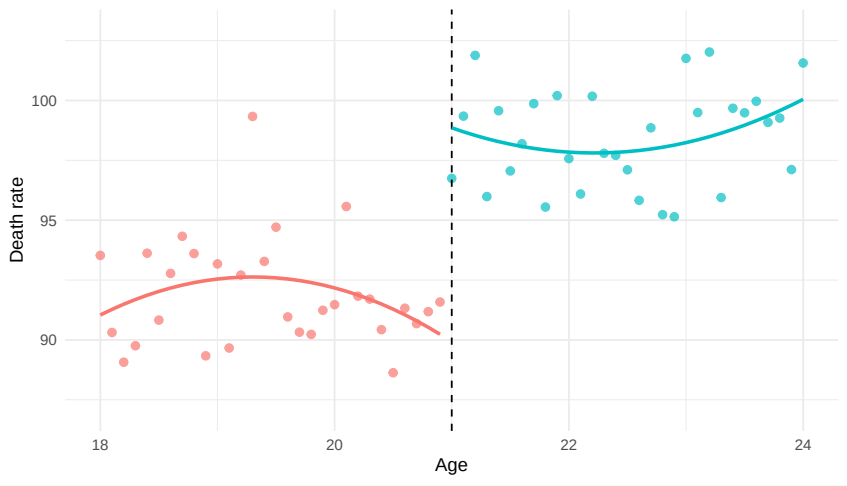
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Bandwidth and Kernel

Bandwidth Choice

We want to give more weight to observations closer to the cutoff level.

Bandwidth

- **Bandwidth** = window around cutoff outside of which you discard observations (e.g. only consider individuals aged 19 to 23; or only those aged 20 to 22)

$\rightarrow$ Narrower bandwidth tradeoff: Assumption credibility v. sample size

Testing A Narrower Bandwidth

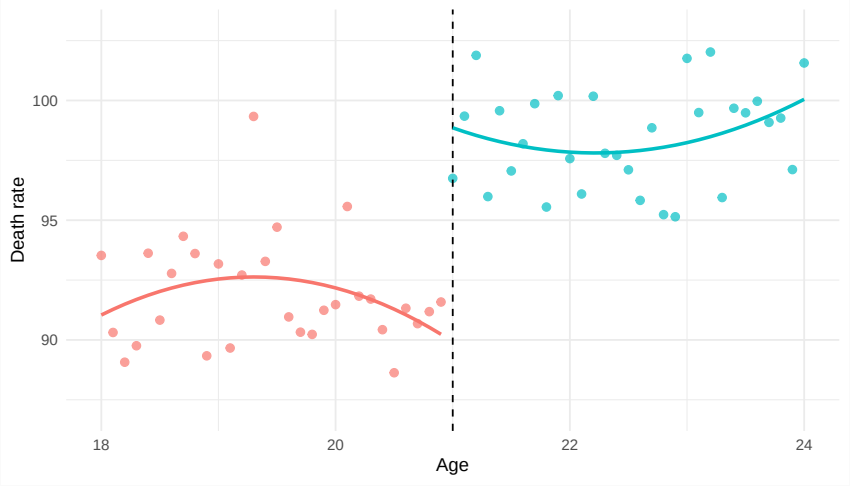
New dataset with only observations within 2 years of the cutoff (19-23):

```
df_narrow <- df %>% filter((age_cell>19) & (age_cell<23))
```

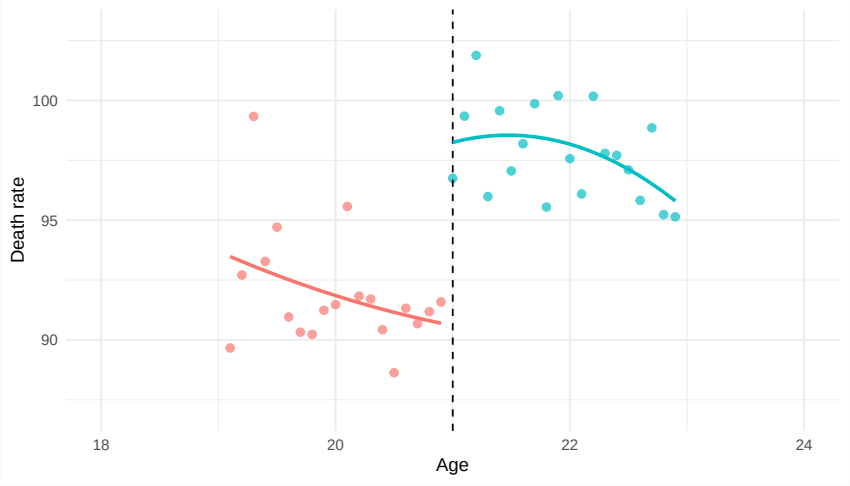
We reproduce the previous specification on the narrower dataset:

```
rdd_quadratic_narrow <- lm(death_rate_per_100T ~ age_centered*treat +
                                                 age_centered_sq*treat,
                           data=df_narrow)
coeftest(rdd_quadratic_narrow, vcov. = vcovHC(rdd_quadratic_narrow, "HC0"))
```

Narrower Bandwidth - Graphical Results



Narrower Bandwidth - Graphical Results



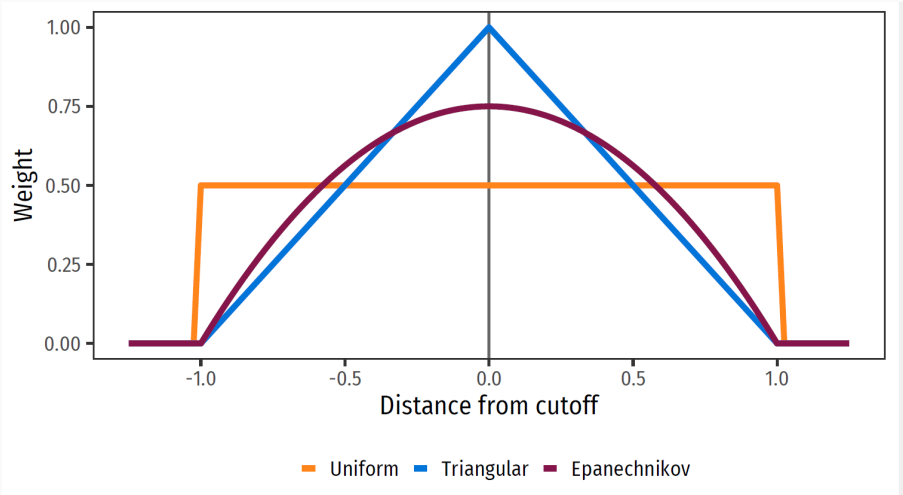
Kernel Choice

We want to give more weight to observations closer to the cutoff level.

Kernel

- **Kernel** = method for assigning importance to values based on their distance to the cutoff
- because we care the most about observations right by the cutoff, we typically want to give more distant ones less weight

Different Kernels



Source: Andrew Heiss

Recap - RDD Implementation

- Recall that implementing RDD requires the research to define:
  1. Functional form of the running covariate.
  2. Bandwidth choice
  3. Kernel weighting

$\rightarrow$ Different choices can lead to different results! Try them all for robustness.